NetCHB Customs Entry XML Preparation & Review System

Candidate: Manthan Khetade • **Role:** Software Engineering Intern • **Topic:** Architecture, Approach & Future Work

1. System Architecture & End-to-End Workflow

We designed a deterministic, TypeScript-native customs extraction and review workflow converting commercial shipping documents (Commercial Invoices, Packing Lists, Bills of Lading) into CBP-compliant NetCHB uploadEntry XML:

- **Multimodal Extraction & Strict Provenance:** We leverage Google Gemini 3.8 Flash via native structured JSON mode to transcribe document observations. Every field adheres to an immutable schema: {value, status, source: {doc, page, snippet}, confidence}. An on-disk cache keyed by file hash, prompt version, and model parameters guarantees instant, zero-cost cache hits on repeat executions.
- **Deterministic XML Serializer:** The LLM *never* generates XML directly. Observations and explicit broker choices feed a deterministic serializer validated against official NetCHB XSDs (entry.xsd & data_type.xsd) using WebAssembly xmllint-wasm. Integer-cents arithmetic prevents floating-point inaccuracies.
- **Dual Validation Architecture:** Structural schema conformity (Schema Valid: PASS/FAIL) is decoupled from statutory filing completeness (Ready to Transmit: N Blockers), ensuring transparent regulatory governance.

2. Generalization, Data Integrity & Customs Discrepancies

To ensure the engine generalizes to unseen commercial shipments without hardcoded logic:

- **Authoritative Pinned Reference Tables:** Integrated official datasets including Census Schedule D (429 U.S. ports with strict role isolation for unlading, entry, and processing ports), Schedule K (2,191 foreign ports), ISO 3166-1 (249 countries), and 82 statistical units derived directly from data_type.xsd.
- **Statutory Discrepancy Reconciliation:** Surfaced real-world document conflicts as actionable broker decision cards rather than silently guessing: invoice math errors ($6,480 printed vs computed), packing shortages (1,176 packed vs 1,200 invoiced), buyer-supplied fabric assists ($4,180), and handwritten container strike-throughs (OPLU3041728 → OPLU3041722).
- **Cryptographic Scoped Invalidation:** Result fingerprints bind lookups to underlying document evidence. Editing an observation invalidates only its applicable confirmation without resetting unrelated broker decisions.

3. Interactive Split-Pane Review Workspace

We developed a responsive React 18 / Vite workspace featuring a split-pane layout: form fields appear alongside verified PDF evidence snippets and a local PDF viewer with hash-matched re-attachment. Actionable issues appear inline next to relevant inputs across 6 filing sections. Brokers can export a local session.json checkpoint to resume reviews later without re-running document extraction.

4. What I Would Prioritize With More Time

- **Multi-Commodity Accuracy Benchmarking:** Expand test suites with diverse commercial shipments (electronics, footwear, machinery, chemical drums) to evaluate table extraction resilience across multilingual document formats.
- **Automated Classification & PGA Integration:** Connect full 10-digit HTSUS subheadings with duty rate inheritance, Section 301/232 trade remedies, Partner Government Agency flags (FDA, EPA, DOT), and automated Manufacturer ID (MID) derivation.
- **Multi-Modal Profiles:** Expand beyond ocean formal consumption (Entry Type 01, Modes 10/11) to air cargo (40), cross-border trucking (30), bonded warehouses (11), and Foreign Trade Zones (21).
- **Production Security & Direct EDI:** Implement encrypted document storage, retention compliance, and direct transmission to NetCHB APIs / CBP ACE test environments.

Repository: github.com/wtf-manthan/sefer-netchb-entry • 123 automated test suites passing • Full coverage across shared, server, and web workspaces.